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Note: The last article is a replacement of authorship, published in *Tech.J.RiverRes.Inst.*: 13(1): 11-25, October 2016, without any changes to the original content (* and **).

Question1:

Write a Python program that takes a user's age as input, checks whether it's a valid number, and prints:

- "You are an adult" if age ≥ 18
- "You are a minor" if age < 18
- "Invalid input" if the input is not a number

Example Output:

Enter your age: 21

You are an adult

Question 2: List and Dictionary Manipulation**Question:**

You have the following list of student names and their marks:

```
students = [("Alice", 85), ("Bob", 72), ("Charlie", 90)]
```

Write a program to:

1. Convert this list into a dictionary.
2. Add a new student ("David", 78) to it.
3. Print the names of students who scored above 80.

Question : OOP — Class and Object**Question3:**

Create a class called Car with attributes brand, model, and year.

Add a method car_info() that prints the full car information.

Then, create two objects of the class and call the method for each.

Example Output:

Brand: Toyota, Model: Corolla, Year: 2020

Brand: Honda, Model: Civic, Year: 2022

Question4:

Create a base class called Animal with a method sound().

Create two subclasses Dog and Cat that override the sound() method.

Then write a function make_sound(animal) that calls the sound() method of any given animal object.

Expected Output:

Dog says: Woof!

Cat says: Meow!

Irrigation Challenges and Opportunities in the Selected Polders of Coastal Bangladesh

M. M. Rahman^{1*}

Abstract

The agricultural sector in Bangladesh, crucial for economic stability and food security, faces significant climate change-induced challenges, especially in coastal polders. This study examines irrigation challenges and opportunities in these regions, focusing on rising food demand and climate change impacts. The construction of 139 polders in the 1960s initially boosted productivity but later caused river siltation, waterlogging, and reduced freshwater availability. Using system dynamics modeling, this research evaluates irrigation demands and solutions under various scenarios, including salinity-tolerant crops and policy interventions like canal re-excavation and groundwater management. Findings reveal disparities in water availability across six selected polders. Monsoon rainfall supports irrigation, but dry-season Boro cultivation is constrained by high surface water salinity. Groundwater is largely unsuitable, except in Polder 29 and Polder 17/1, where salinity remains within limits. Future projections indicate increased irrigation demands, particularly for Polder 29 and Polder 17/1, which can be met through sustainable groundwater use and canal rehabilitation. In contrast, polders like 35/1 and 46 require canal re-excavation to enhance water storage. Rising sea levels and soil salinity are expected to reduce cultivable land, though salt-tolerant crops may partially offset losses. Policy interventions prioritizing surface water management, including canal excavation, are essential to mitigating water scarcity. This research highlights the need for integrated water resource management and climate adaptation strategies, offering insights to guide sustainable irrigation practices in Bangladesh's vulnerable coastal regions.

Keywords: Cause and loop diagram, Climate change, Coastal zone, Groundwater, Irrigation, Surface water, Stock flow diagram, System dynamics (SD) model.

Introduction

Agriculture is one of the most paramount sectors of the economy of Bangladesh together with ready-made garments (RMG), remittances, and others (Islam and Haque, 2018; Rahman, 2017). Traditionally, Bangladesh is an agricultural country characterized by fertile land and a favorable climate for crop production, with approximately 83% of the rural population relying on agriculture for their livelihoods. Of the cultivable land, 28% is dedicated to single cropping, 49% to double cropping, and 22% to triple cropping, while 5% remains fallow (BBS, 2018). However, the sector faces significant challenges, including rising food demand from a growing population and the decline of agricultural land due to urbanization and hydro-climatic hazards such as salinity and waterlogging (IWWI, 2015; Mainuddin and Kirby, 2015). Bangladesh is particularly vulnerable to climate change, experiencing severe impacts like sea-level rise, salinity and storm surge which threaten agricultural productivity, especially in the low-lying coastal areas where 28% of the population resides (Brammer, 2014; Hossain and Majumder, 2018; Xiaoying and Sikder, 2014). Recent studies have highlighted that soil and water salinity levels in coastal areas such as Assasuni, Dacope, and Morrelganj have continued to rise, particularly affecting irrigation availability in the dry season (Shapna et al., 2024). These issues result in lower crop intensity in the southwest coastal zone compared to the national average (Rashid et al., 2018). Additionally, rising temperatures due to climate change onward to higher evapotranspiration rates, increasing the water requirements for agriculture (Dang et al., 2024). While agricultural activities in the

wet season primarily rely on rainwater, dry season crops depend heavily on irrigation. However, elevated salinity levels render surface water unsuitable for irrigation during this period.

Challenges in Polder Irrigation Water Management

To make the land productive, the government of Bangladesh constructed 139 large-scale polders with the help of donors, occupying 1.2 million hectares and involving 5,107 km of embankments, water control structures, drainage canals, and storage channels in the 1960s and 1970s. Polders were constructed in the coastal zone to protect communities from tidal flooding, and salinity intrusion, and make agricultural land more productive (Islam and Rahman, 2015; Rahman and Rahman, 2015). The polders had an immediate and significant impact on agricultural output, which increased in the first 10-15 years (Nowreen et al., 2014). This infrastructure enabled greater access to markets for trading goods, resulting in a boost in income and standard of living for people.

However, after a few decades, negative consequences of the polders have risen (Deb, 1998; Paul et al., 2013; Rahman et al., 2010). Polder construction has significantly altered the sediment balance in the delta, inducing subsidence and waterlogging in some of the polders (Ahmed, 2011; Islam et al., 2002). The intrusion of storm surge saline water causes long-term problems for crops and drinking water sources. After a storm surge, a long time is needed to drain out saline water due to the low elevation difference between the river and the inside polder. Moreover, the lack of proper functioning of water control structures (sluice

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gates, inlets, outlets) allows saline water to intrude during the dry season, polluting stored freshwater in canals intended for irrigation and domestic use. Over time, canals have become silted up due to poor maintenance and topsoil erosion from nearby land (Ahsan et. al., 2022, Shawkhatuzamman et. al., 2023).

The present research highlights the growing threats of salinization and waterlogging in Dutch polders, which are driven by climate change, sea-level rise, and intensified agricultural activity. Van Der Burg et. al., (2024) outline five key adaptation strategies to maintain agricultural productivity under saline conditions: development of salt-tolerant crop varieties, cultivation of halophytes, application of soil management techniques, use of bio stimulants, and implementation of advanced irrigation systems. These strategies are assessed for sustainability across environmental, social, and economic dimensions. Complementary to these findings, Ritzema and Stuyt (2014) discuss land drainage strategies to cope with climate change in the Netherlands, emphasizing the importance of controlled drainage systems to manage water levels and mitigate salinization in polders.

Rainfall is generally sufficient to meet the irrigation water demand for the Kharif I and Kharif II seasons, with only a small amount of irrigation needed during the transplanting stage of Kharif II. This demand is typically satisfied using surface water, as salinity levels are acceptable for irrigation during the Kharif II or monsoon season. However, dry season crops primarily depend on irrigation, and raised salinity levels during this period reduce surface water unsuitable for agricultural use. Consequently, the lack of irrigation water restricts crop cultivation in the dry season within the coastal zones of Bangladesh. Groundwater in most polders in the southwest coastal region is also unsuitable for irrigation and domestic purposes due to high salinity, and failing to meet drinking water standards (Bahar and Reza, 2010). As a result, irrigation in these polders relies heavily on surface water during the dry season. Unfortunately, since the construction of the Farakka Barrage, surface water salinity has increased due to reduced upstream flow during dry periods (Mahtab and Zahid, 2018). Furthermore, high salinity in irrigation water significantly reduces yields for Boro and Aman paddy (Khanom and Salehin, 2012). However, research indicates that improving canal systems for storing fresh water, such as rainwater, could enhance crop production and increase cultivated land (Ahmed and Islam, 2013; Begum et al., 2024; Sharifullah et al., 2008). Another significant challenge for agriculture in the polder areas is the increasing salinity caused by rising sea levels, which reduces the amount of cultivable land.

In light of these challenges, the present study aims to assess future irrigation water demand under various plausible scenarios and explore potential irrigation

options for dry season Boro crops in selected polders of the southwest coastal zones of Bangladesh using a system dynamics approach.

Methodology

Study area selection

For this study, Polder 17/1, Polder 29, Polder 32, Polder 35/1, Polder 43/2F, and Polder 46 from the southwest and south-central regions of coastal Bangladesh were selected to investigate irrigation water availability for Boro and rabi crops during the dry season (**Fig. 1**). Additionally, plausible options and interventions for improving irrigation water availability were analyzed in light of climate change scenarios.

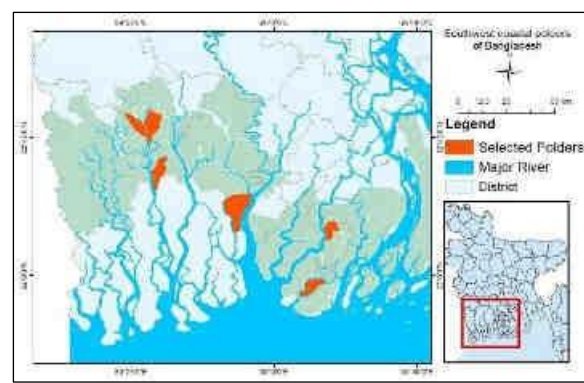


Fig. 1. Selected study polders

Data collection

Secondary data on rainfall, groundwater, water control structures, polder canals, water quality, agriculture, and irrigation were collected from sources like BMD, BWDB, BBS, and various reports. These included rainfall records, groundwater depth, canal specifications, water quality metrics, and agricultural statistics. Additionally, primary data were gathered through Key informant interviews (KIIs), focus group discussions (FGDs) and group discussions (GDs), and interviews to understand local perceptions of irrigation water scarcity, assess cultivated land, and explore potential solutions. Union Parishad members and Upazila agriculture officers provided insights on Boro cultivation areas, irrigation sources, and related challenges.

Model Development

The System Dynamics (SD) model integrates both qualitative/conceptual and quantitative/numerical modeling methods. Qualitative modeling, such as using a causal loop diagram (CLD), enhances our conceptual understanding of the system. Quantitative modeling, such as employing a stock-and-flow diagram (SFD), allows us to investigate and visualize the effects of

different intervention strategies through simulation. The model incorporates biophysical and socioeconomic elements, including surface water (SW), groundwater (GW), rainfall, water control structures, crop water requirements, population, and more. These elements are interconnected through 'cause and effect' relationships. Detailed descriptions of each modeling step are provided in the following sections. The model was developed using the Vensim DSS tool by Ventana Systems Inc., Harvard, Massachusetts, USA (Ventana Systems, 2019).

Causal Loop Diagram for qualitative model development

In a causal loop diagram (CLD), system elements are interconnected by arrows that link cause-and-effect variables to form causal chains. **Fig. 2** illustrates the

causal loop diagram, featuring positive (+) and negative (-) signs to indicate the nature of these relationships. A positive sign denotes a direct relationship between variables: when the causative variable increases, the effect variable also increases, and conversely, a decrease in the causative variable results in a decrease in the affected variable. For example, an increase in population leads to higher food requirements, which in turn drives an increase in cultivated land within a tolerable threshold, assuming other variables remain constant. In contrast, a negative sign indicates an inverse relationship between the variables; for instance, increased groundwater consumption results in a decrease in groundwater levels. Feedback loops are a crucial component of the system dynamics model, and they can be categorized into two types: balancing loops (B) and reinforcing loops (R).

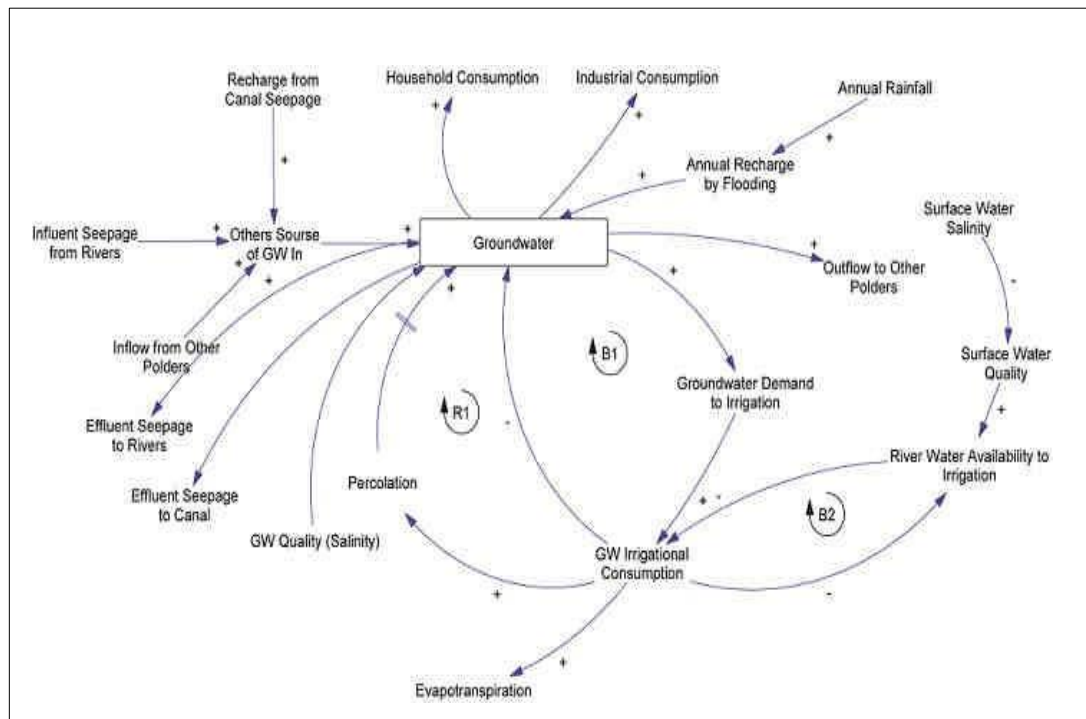


Fig. 2(a). Groundwater component with irrigation water demand.

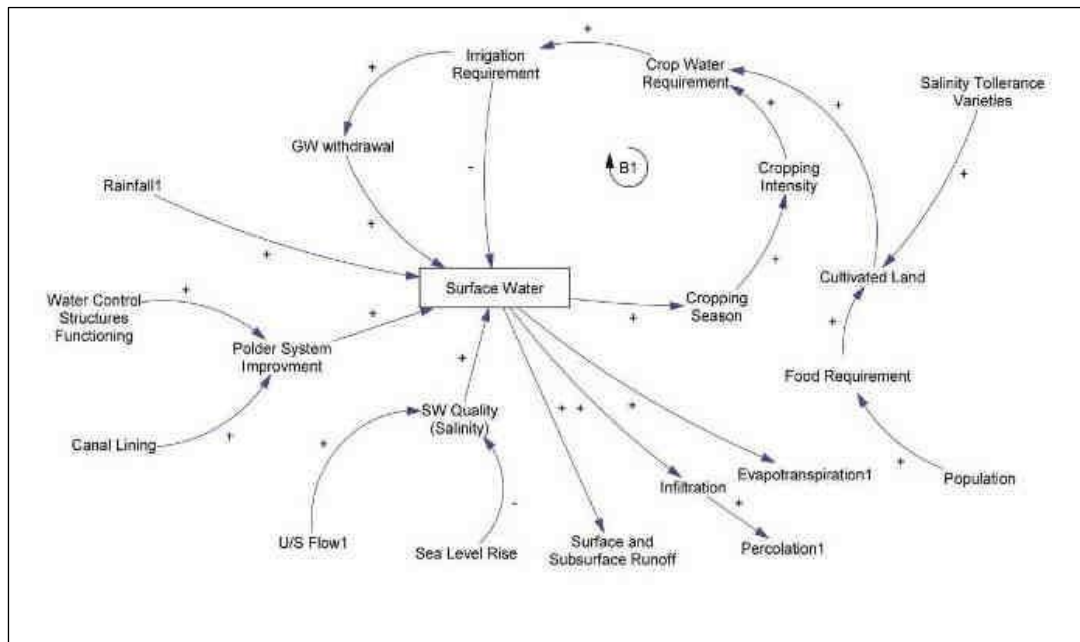


Fig. 2(b). Surface water component with irrigation water demand.

Fig. 2. CLD of the study model. Arcs describe the directions of influence. A positive arc reads as “an increase in variable A leads to an increase in variable B”. A negative arc reads as “an increase in variable A leads to a decrease in variable B”.

Stock Flow Diagram for Quantitative Model Development

A causal loop diagram was used to visually represent the system, while a stock-flow diagram was used for quantitative analysis. This included deterministic, stochastic, and empirical equations. The stock-flow diagram, shown in **Fig. 3** illustrates how rainfall, water volume, pluvial flooding, and crop losses are connected. The detailed mathematical equations are explained in the following sections.

Surface and groundwater

Groundwater volume at any time can be computed by

Eq.

$$Stock(t) = \int_0^{t_n} [GW(t) + GR(t) - I(t)]dt + GWL(t_0) \quad \text{Eq. (1)}$$

Where, $GW(t)$ = Groundwater level, $GR(t)$ = Groundwater recharge, $I(t)$ = Irrigation water withdrawal from groundwater, $GWL(t_0)$ = Groundwater level t_0

Groundwater, surface water at any time can be computed by Eq. (2):

$$Stock(t) = \int_0^{t_n} [SW(t) + SWI(t) - SWO(t)]dt + SWL(t_0) \quad \text{Eq. (2)}$$

Where, $SW(t)$ = Surface water volume, $SWI(t)$ = Surface water inflow, $SWO(t)$ = Surface water outflow

$SWL(t_0)$ = Surface water volume t_0

Population

Projection of population of polders logistic growth method has been implied. If P_0 , P_1 and P_2 are the populations of an area at time t_0 , t_1 and t_2 , the saturation population equivalent to the carrying capacity of the area can be calculated using the following equation.

Saturation population, P_s

$$2P_0P_1P_2 - [P_1 \times P_1(P_0 + P_2)] = \frac{(P_0P_2 - P_1P_1)}{0 \quad 2 \quad 1 \quad 1} \quad \text{Eq. (3)}$$

Sea Level Rise

This study assumes a linear increase in the sea level over time, based on IPCC projection. The SL, at a given time, is calculated by:

$$SL_t = \int dRdt + SL_0 \quad \text{Eq. (4)}$$

Where SL_t is the linear sea level at time t , dR is the rate of rising at each time step dt , and SL_0 is the initial sea level at the beginning of the simulation. According to the IPCC 5th assessment report, in the GBM delta, sea level may rise 0.63 to 0.88 m by 2090 (Kay et al., 2015).

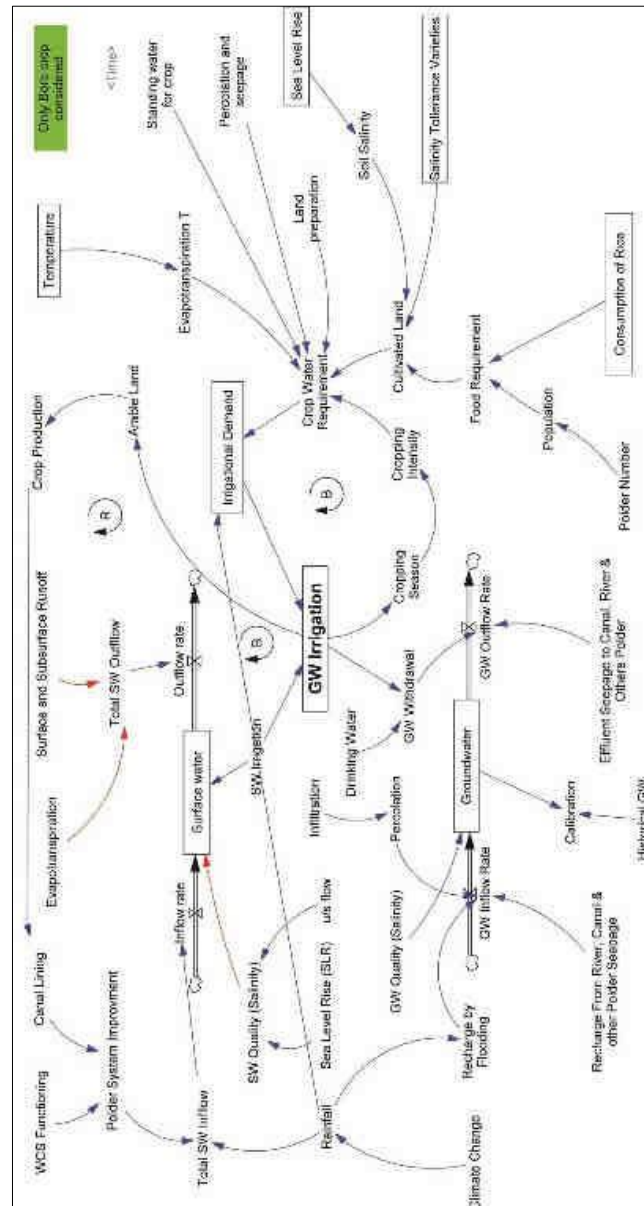


Fig. 3. SFD of the study model. Boxed variables represent stocks, double arrows represent flows that increase or decrease stock levels.

Groundwater quality

Detailed groundwater scenarios of selected polders are shown in **Table 1**.

Table 1. Different aquifers' salinity level (dry season), March 2012 (Data Source: IWM, CEIP, Blue Gold).

Polder No	1 st (upper shallow) aquifer (up to 30m) EC ($\mu\text{S}\cdot\text{cm}^{-1}$)	1 st (lower shallow) aquifer (up to 100m) EC ($\mu\text{S}\cdot\text{cm}^{-1}$)	2 nd Aquifer (above 100m) EC ($\mu\text{S}\cdot\text{cm}^{-1}$)	Remarks
Polder 17	1000-3000	<2000	<1000	usable
Polder 29	1000-3000	<2000	<1000	usable
Polder 32	8000-9000	6000-7000	4000-5000	Not usable
Polder 35	13000-14000	8000-9000	7000-8000	Not usable
Polder 43/2F	14000	13600	941	Not usable
Polder 46	27400	30000	21000	Not usable

Surface water salinity

Table 2 shows the salinity level of different polders of different coastal areas.

Table 2. Different polders surface water salinity level (Data Source: CEIP, Blue Gold, BWDB).

Polder Name	River	District	Upazila	Date of sampling	EC ($\mu\text{S-cm}^{-1}$)
Polder 29	Bhadra, Haparkhali	Khulna	Dumuria, Batiaghata	11.11.2012	323
				08/12/2012	1710
				20/01/2013	3060
Polder 17/1	Teliganga-Gengrail	Khulna	Dumuria	11.11.2012	475
Polder 32	Shibsa, Bhadra, Chunkuri, Sutarkhali	Khulna	Dacope	08/11/2012	1790
				15/12/2012	2510
				20/01/2013	4850
Polder 35/1	Baleswer, Bhola, Sannasir Khal	Bagerhat	Sarankhola	29.11.2012	1415
				22.12.2012	1680
				27.10.2013	2080
Polder 43/2F	Gulishakhali, Payra, Kukua	Barguna	Amtali	06.10.2012	203
				10.12.2012	315
				21.02.2012	580
Polder 46	Tiakhali, Nilakhi	Patuakhali	Kalapara	05.10.2012	441

Canal storage

The present canal capacity of selected polders, design capacity and dry season capacity are shown in **Table 3**.

Table 3. The length and the capacity of canals of selected polders (Data sources: Blue Gold, CEIP, BWDB).

Polder No	Length (km)	Design Capacity (Mm^3)	Present capacity (60% of design capacity) (Mm^3)	Present capacity during monsoon (60% of present capacity) (Mm^3)	Design Capacity during Monsoon (60% of design capacity) (Mm^3)
Polder 29	157.00	6.62	3.97	2.38	3.97
Polder 43/2F	93.50	2.63	1.58	0.95	1.58
Polder 32	70.00	4.30	2.58	1.55	2.58
Polder 35/1	251.26	7.08	4.25	2.55	4.25
Polder 17/1	58.00	3.14	1.88	1.13	2.35
Polder 46	71.84	1.43	0.86	0.51	0.86

Irrigation water requirements

The FAO model was used to determine the irrigation water requirement of the Boro crop. According to this model, total irrigation water requirement can be calculated as,

$$W_{irr} = ET_{crop} + W_{lp} + W_{ps} + W_l - P_e \quad \text{Eq. (5)}$$

Where, W_{irr} – Irrigation water requirements

ET_{crop} – Crop evapotranspiration

W_{lp} – Water required for land preparation

W_{ps} – Percolation and seepage losses of water from the field

W_l – Water to establish a standing water layer

P_e – Effective precipitation

Crop evapotranspiration is calculated as,

$$ET_{crop} = K_c * ET_{ref} \quad \text{Eq. (6)}$$

Where K_c is the crop coefficient, and ET_{ref} is the reference evapotranspiration. The Penman-Monteith

method was used to calculate reference evapotranspiration from climate data.

Results and Discussion

Model verification

The model verification results showed a strong alignment between simulated and historical Boro cultivated land percentages. In Polder 29, the model estimated 22.0% cultivation in 2008, while historical data ranged from 13.0% to 21.0%. For 2015, the model predicted 23.8%, closely matching the historical 23.3%. Similarly, in Polder 17/1, the model estimated 21.7% in 2008, with historical data ranging from 12.0% to 22.0%, and for 2015, the model projected 23.0% compared to the observed 23.4%. These results confirm the model's accuracy in simulating Boro cultivation trends.

Scenario development

Three scenarios and two policies were considered in the study. The first scenario, "Business as Usual," assumes that Boro-cultivated land will expand with increasing food demand while other factors remain unchanged. The second scenario incorporates climate change impacts, including rising temperatures (Increased 4°C by 2100; Fahad et al., 2018), sea-level rise (0.63 to 0.88 m by 2090; Kay et al., 2015), and rainfall increased -8.22 % in dry by 2080 for southwest (SW); Fahad et al., 2018. The third scenario explores the use of salinity-tolerant crop varieties to adapt to changing conditions. In terms of policy interventions, two options were evaluated: surface water management through re-excavation and excavation of polder internal canals, and groundwater management by installing tube wells to enhance irrigation capacity.

The present condition of irrigation

Rainfall typically provides enough water to meet irrigation needs during the Kharif I and Kharif II seasons. However, a small amount of additional irrigation is required during the transplanting stage of Kharif II. This irrigation is generally supplied using surface water, as its salinity levels remain suitable for

irrigation throughout the Kharif II or monsoon season. However, dry season crops, particularly Boro rice, primarily rely on irrigation, and higher salinity levels during this period reduce surface water unsuitable for agricultural use. Groundwater in most polders is also not viable for irrigation, leading to significant fallow land during the dry season. Farmers in some areas of the selected polders cultivate sesame, vegetables, and Boro rice during this time. **Table 4** presents the present cultivated land, irrigation water demand, and sources of irrigation. The table indicates that irrigation water is not required for sesame and vegetables. Polder 29 and Polder 17/1 have higher cultivated land areas compared to other polders, and their irrigation demand is also greater. Surface water is the primary source of irrigation for all polders, except for Polder 29 and Polder 17/1, which predominantly depend on groundwater for irrigation, with only 20% of their irrigation sourced from surface water. Polder 43/2F does not require irrigation water as there is no Boro cultivation due to socio-economic factors. Although this polder has access to both groundwater and surface water sources, farmers do not cultivate Boro rice, as it is considered less profitable than other crops. Farmers primarily use shallow tube wells for groundwater irrigation and low-lift pumps (LLPs) for surface water irrigation.

Table 4. Present condition of dry season irrigation (estimated).

Polder No	Boro cultivated land (ha)	Sesame and Vegetables cultivated (ha)	Water requirement (mm)	Water Used (Mm ³)	Type of irrigation	GW used (Mm ³)	SW used (Mm ³)
P29	1273		1220	15.53	Groundwater irrigation is provided in 80% of areas, the remaining portion is provided by LLPs	12.20	3.33
		219	300	0.66	No irrigation is required	-	-
P17/1	935		1220	11.41	Groundwater irrigation is provided in 80% of areas, the remaining portion is provided by LLPs	9.13	2.28
		177	300	0.53	No irrigation is required	-	-
P43/2F			1220	0.00	Surface water irrigation	-	-
		623	300	1.87	No irrigation is required	-	-
P32	300		1220	3.66	Surface water irrigation	-	3.66
		270	300	0.81	No irrigation is required	-	-
P35/1	225		1220	2.75	Surface water irrigation	-	2.75
		4910	300	14.73	No irrigation is required	-	-
P46	28.88		1220	0.35	Surface water irrigation	-	0.35
		1917	300	5.75	No irrigation is required	-	-

Future Boro cropland under different scenarios

The scarcity of quality irrigation water during the dry season limits the cultivation of Boro rice and other crops. Despite these challenges, some farmers manage to grow Boro rice, sesame, and vegetables, while many others leave their land fallow. This study simulates future Boro rice cultivation under a business-as-usual (BAU) scenario, which considers the increasing food demand from the growing population without accounting for factors such as climate change, sea-level rise, soil salinity, or the introduction of salt-tolerant varieties (STVs). In the BAU scenario, it is assumed that the area dedicated to Boro crops will increase in response to rising food demand, while the impacts of climate change and salt-tolerant varieties are not

factored in. Conversely, the climate change (CC) scenario anticipates a decrease in Boro cropland due to the exacerbation of soil salinity caused by climate change and rising sea levels. Additionally, the introduction of salt-tolerant varieties is considered in a separate scenario, which represents two competing situations: an increase in Boro cropland due to the availability of improved crop varieties, and a decrease in Boro cropland resulting from climate change-induced soil salinity. **Fig. 4** shows future simulated Boro cropland under different scenarios for six selected polders. The figure shows that Boro rice land will increase in the BAU scenario for all polders, but the land will decrease in the climate change scenario. Instead of a negative impact, salt tolerant variety will positively impact Boro rice cropland.

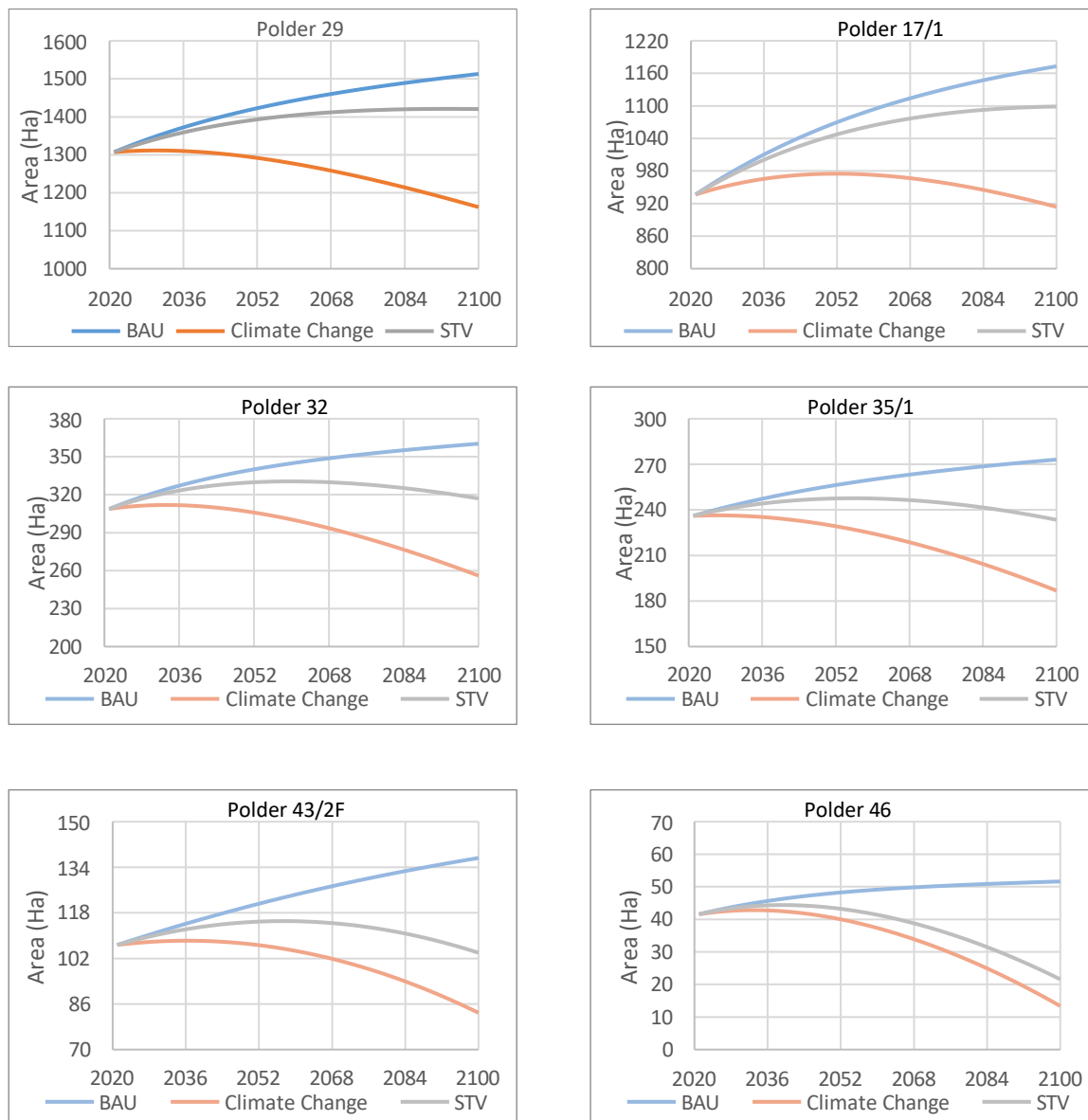


Fig. 4. Boro cropland areas under business as usual (BAU), climate change, salt tolerant crop variety (STV) scenarios.

Fig. 5 illustrates the future simulated water demand for Boro crops under three scenarios: business-as-usual (BAU), climate change, and salt-tolerant crop variety (STV). In the BAU scenario, water demand is projected to increase as Boro cultivation expands to meet the growing population's food demand. However, in the climate change scenario, Boro water demand is expected to decrease due to increasing soil salinity, which will ultimately reduce the land available for Boro cultivation. The introduction of salt-tolerant crop varieties is expected to increase Boro water demand, as more land will come under cultivation. Changes in

evapotranspiration due to rising temperatures will also impact water demand, although the effect is not significant. Boro crop water demand is higher in Polder 29 and Polder 17/1 because more land will be cultivated due to lower soil salinity levels. In Polder 43/2F, the change in Boro water requirement is not significant because people are not interested in Boro cultivation due to socioeconomic factors. In the other polders, Boro crop water demand is not as high because more land will not come under cultivation due to higher soil salinity levels.

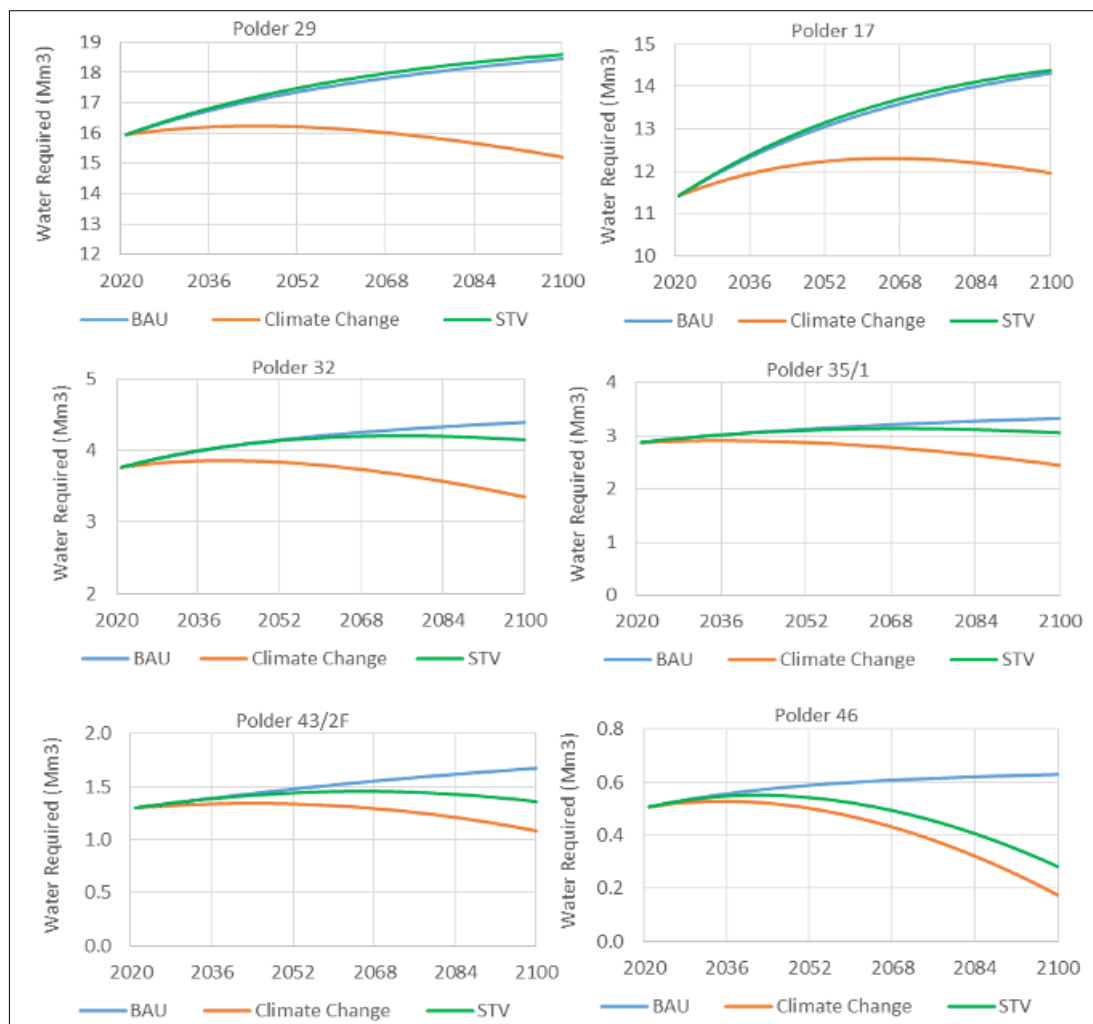


Fig. 5. Boro water requirement under business as usual (BAU), climate change, salt tolerant crop variety (STV) scenarios.

Crop water versus irrigation water

Fig. 6 illustrates the total crop water requirement and irrigation water demand. While rainfall during the dry

season is limited, it still contributes to meeting crop water needs alongside irrigation. Boro rice, however, relies primarily on irrigation water to satisfy its water requirements.

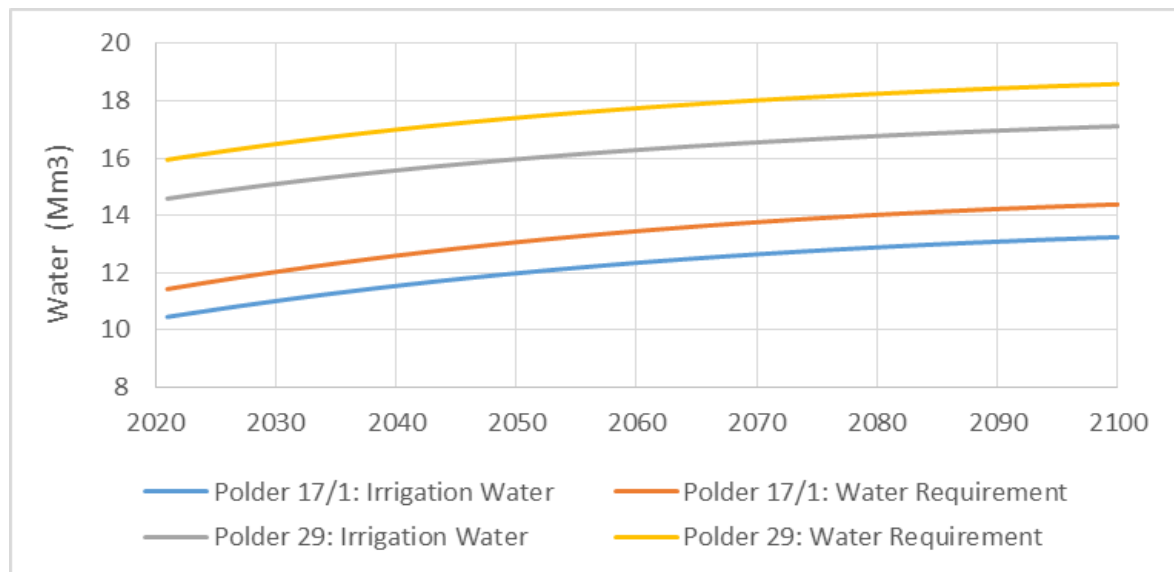


Fig. 6. Crop water requirement versus irrigation water requirement

Surface and groundwater availability for irrigation

Table 5 presents the current and future irrigation water demand and sources for six selected polders. It indicates that Polder 29 and Polder 17/1 possess sufficient groundwater potential to expand Boro cultivation. Although Polder 43/2F has access to both groundwater and surface water for irrigation, the future

irrigation water demands of Polder 35/1, Polder 46, and Polder 43/2F can only be met through the re-excavation of existing canals. In contrast, Polder 32 will require the excavation of new canals to satisfy its future irrigation water needs. Overall, re-excavation of canals across all polders is essential to enhance their storage capacity.

Table 5. Future irrigation water, its sources and probable interventions for selected polders.

Polder No.	Year	Irrigation water demand (Mm ³)	Sources of water		Interventions & remarks
			SW (Mm ³) (Present and after re-excavation)	GW (Mm ³)	
P 29	2021	14.59	2.92	11.67	Have groundwater potential
	2050	15.96	4.63	11.33	
	2100	17.11	4.63	12.48	
P 17/1	2021	10.45	2.09	8.36	Have groundwater potential
	2050	11.98	2.20	9.89	
	2100	13.24	2.20	11.15	
P 35/1	2021	2.63	3.54	No need & not usable	Need re-excavation of canal
	2050	2.84	4.96	No need & not usable	
	2100	2.81	4.96	No need & not usable	
P 32	2021	3.44	2.58	Need but not usable	Construction of more canals and reservoirs
	2050	3.78	3.23	Need but not usable	
	2100	3.82	3.23	Need but not usable	
P 43/2F	2021	1.19	1.32	No need but usable	Have the potential to extend Boro rice
	2050	1.32	1.84	No need but usable	
	2100	1.25	1.84	No need but usable	
P 46	2021	0.46	0.71	No need & not usable	Need re-excavation of canal
	2050	0.50	1.01	No need & not usable	
	2100	0.46	1.01	No need & not usable	

Groundwater sustainability

There are two primary concerns regarding groundwater use in the coastal zone of Bangladesh. Excessive pumping leads to a decline in groundwater levels, which may threaten groundwater sustainability. Additionally, increased water withdrawal can cause saline water intrusion into the groundwater supply. Polder 29, Polder 17/1, and Polder 43/2F have the

potential to utilize groundwater effectively, as salinity levels are within acceptable limits for irrigation. **Fig. 7** displays the observed groundwater table at the Dumuria station in Polder 29. The graph indicates a significant recovery of groundwater levels during the monsoon season, following recharge from monsoon rainfall. Furthermore, there has been no significant trend in the groundwater table over the past twenty-two years.

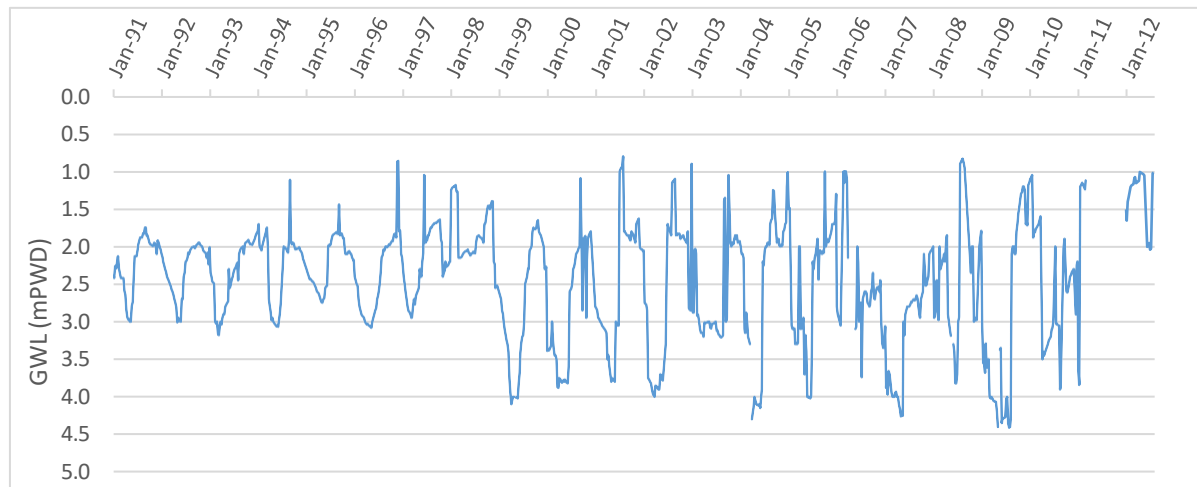


Fig. 7. Temporal variation of groundwater level for Polder 29 (Data source: BWDB).

Conclusion

Climate change-induced salinity is projected to impact Boro rice cultivation and irrigation water demand in coastal Bangladesh. While salinity may reduce Boro cropland, the introduction of salinity-tolerant varieties could increase water demand. Currently, the irrigation water demands for Polder 29, Polder 17/1, Polder 32, Polder 35/1, Polder 43/2F, and Polder 46 are 14.59, 10.45, 2.63, 3.44, 1.19, and 0.46 million cubic meters (Mm³), respectively. By 2050, the demands are projected to rise to 15.96, 11.98, 2.84, 3.78, 1.32, and 0.50 Mm³, respectively. The increase in water demand is substantial only for Polder 29 and Polder 17/1. In terms of water availability, surface and groundwater sources are insufficient to meet the Boro rice cultivation needs. Groundwater in all polders, except Polder 29 and Polder 17/1, is unusable due to high salinity. Although the second aquifer groundwater in Polder 43/2F is usable, national policy restricts its use for irrigation. To address increased water demand, re-excavation of existing canals is needed for Polder 35/1 and Polder 46. Polder 32 requires both new canal excavation and re-excavation of existing canals, as surface water is the only available irrigation source. Polders 29, 17/1, and 43/2F can meet their irrigation water needs through groundwater and canal re-excavation, with no significant groundwater sustainability issues.

Conflict of Interest

The authors declare no conflict of interest.

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ASSESSMENT OF EROSION RISK POTENTIAL IN TERMS OF BANK MATERIAL PARTICLE SIZES AROUND 3KM UPSTREAM AND 3KM DOWNSTREAM OF HAJI SHARIATULLA BRIDGE ALONG ARIAL KHAN RIVER IN BANGLADESH

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Abstract

River bank erosion is a recurring issue at monsoon in Bangladesh. Erosion expedites the river bank failure occurring among many of the river bank failure criteria. So, it is very much important to assess bank erosion of the river prior to the planning of bank protection works. Here an attempt has been made to assess the erosion risk potential at the left and right banks around 6.0 km of Arial Khan River. In this study particle sizes are analyzed and percentage of Clay, Silt and Sand has been found out. From the particle size analysis, it has been seen that the left bank of the Arial Khan River predominantly consists of sand particles, except for Khas Char Bachamara, where silt particles dominate. On the other hand, right bank of Arial Khan River dominates the Silt and Sand particle. The erodibility index is determined through Bouyoucos erodibility index and El_{ROM} . From the erodibility index result of El_{ROM} the soil erosion of the bank is categorized and observed that the left bank of Arial Khan River is very high to critical position in risk of erosion except 3.0 m depth of Khas Char Bachamara which layer sometime protects the bank from erosion. On the other hand, the right bank of Arial Khan River is high to critical position in risk of erosion. From this study a design engineer can be aware of the bank materials and predict erosion risk. However, it is recommended that a full geotechnical investigation is needed to prevent river bank erosion.

Keywords: Arial Khan River, Bank materials, Erosion risk potential, Particle sizes.

Introduction

The major part of Bangladesh is on the delta formed by three significant rivers Brahmaputra, Ganges and Meghna. Originating beyond the country's borders, these rivers, along with numerous smaller ones, constitute the Ganges-Brahmaputra-Meghna River system. A wide delta has been formed over the past thousand years due to sediments from these rivers, which have shaped the vast area of Bangladesh and the submerged delta-plain in the Bay of Bengal. 80% of the country's soil is formed by these sedimentary deposits. The remaining 20% of soils, 12% are obtained from Tertiary and Quaternary sediments found in hills, while 8% are uplifted Pleistocene terraces. (Banglapedia, 2021). River bank erosion is a frequent occurrence during monsoon in Bangladesh. Almost every year river banks are facing problems like erosion and failure occurs due to erosion along with other river bank failure criteria (Alom *et al.*, 2020). By this way bank materials play an important role on river bank erosion as well as failure.

The Arial Khan river is a Prominent and highly active river in Bangladesh, creates significant bank erosion which is one of the main reasons for the suffering of the local people (Akter *et al.*, 2013). This river serves as a significant branch of the Padma river's distributary network. It originates from the Padma river at 51.5 km downstream of Goalanda of Rajbari district. The river runs through Faridpur, Madaripur and Barishal districts and maintains meandering channel through its course and is erosion in nature. The river is navigable round the year and has tidal characteristics. Its normal tidal range at Madaripur is 0.32 m. This river covers a total length of 160.0 km (Hossain *et al.*, 2007). Severe river bank erosion have caused destruction of number of settlements in vicinity of bank. Under such circumstances a study has been undertaken in order to assessment of the bank erosion in terms of particle sizes of bank material around 6.0 km of the left bank and right bank of Arial Khan River at

Shibchar Upazilla under Madaripur district of Bangladesh. Watson *et al.* (2006) studied stream bank erosion with a review of processes of bank failure, measurement and assessment techniques and modelling approaches. They stated that stream bank erosion depends mainly on flow pattern, bank material composition, bank geometry, channel geometry, bank soil-moisture conditions, vegetation and man-induced factors. Talukdar (2006) studied the probable causes of erosion, river instability, soil erosion, sediment transport and deposition. He concluded that sediment gets into the river not only from the catchment area but is also contributed by erosion of its banks. He defined that causes of failure of river banks can be directed erosion of the river bank among other probable causes. Roslan *et al.* (2013) worked on river bank erosion erodibility of Langat river of Malaysia and found the Sheet, Rill and Gully three types of soil erosion features commonly seen along Langat river. They found that along the Langat river the soil classifications predominantly consist of sand and silt, commonly referred to as well-graded silty sand, sandy clay of low plasticity and very clayey sand. They stated that soils with a greater proportion of sand show a greater risk of failure which is substantiated and affirmed by the 'ROM' scale equation where most of the soil samples were categorized as very high and critical in the degree level of erosion risk and thus will lead to risk of river bank failure. 'ROM' scale (after the name of the researchers Roslan and Mazidah) is used to ascertain the degree of soil erodibility namely low, moderate, high, very high and critical along the river bank based on the percentage composition of sand, silt and clay. They concluded that the responsible authority could implement proactive measures to prevent adverse incidents along the Langat river at an early stage.

The soil scientist Bouyoucos invented Eq. (1) for determining the Bouyoucos erodibility index (BEI) on the basis of composition of soil

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$$BEI = \frac{\% \text{ Sand} + \% \text{ Silt}}{\% \text{ Clay}} \quad \text{Eq. (1)}$$

This lack of versatility has led to the development of an advanced and new improved soil erodibility index in the name of EI_{ROM} or 'ROM' Scale as in Eq. (2).

$$EI_{ROM} \text{ equation} = \frac{\% \text{ Sand} + \% \text{ Silt}}{2 \times (\% \text{ Clay})} \quad \text{Eq. (2)}$$

This new equation still uses the original principle of Bouyoucos, which is to analyze the soil textural composition of sand, silt and clay. The new equation has effectively demonstrated both the significant value and the threshold for delineating soil erodibility, while also indicating the anticipated erosion characteristics. With the new EI_{ROM} equation outlined in Eq. (2), the more realistic and significant value of soil erodibility index can be used concurrently considering its corresponding risk category as illustrated in **Table 1** to denote the extent of soil erodibility.

Table 1. 'ROM' scale with regards to soil erodibility category.

'ROM' scale	Soil erodibility category
< 1.5 Low	Low
1.5 -4.0	Moderate
4.0 -8.0	High
8.0 -12.0	Very High
> 12.0 Critical	Critical

(Source: Roslan and Mazidah, 2012)

$EI_{ROM} = (\% \text{ sand} + \% \text{ silt}) / 2(\% \text{ clay})$ The 'ROM' scale was developed solely on the grading characteristics of soil, utilizing the EI_{ROM} equation to derive the soil erodibility index and ascertain the configuration of erosion features. In other words, the 'ROM' scale quantifies the degree of soil erodibility by applying the EI_{ROM} equation. This pioneering scale represents the first endeavor to categorize erosion severity based on both the soil erodibility index and erosion features. During the early stages of developing the 'ROM' scale, various areas and locations across Peninsular Malaysia experiencing erosion-induced landslides were pinpointed. Physical reconnaissance and observation of soil erosion features, coupled with basic soil classification information, were documented during surveys conducted in the identified areas. The impact of the soil erosion process on the affected areas recognized from the erosion features commonly known as sheet, rill, and gully were critically observed. The identification of the soil textural composition namely sand, silt and clay are crucial in determining the risk of landslides caused by erosion. The EI_{ROM} equation as stated in advance was created to gain an acceptable sound erodibility value

relative to other scale of measurements. Duong *et al.* (2019) studied the processes of river bank failure along the red river bank using three distinct bank soil and by modeling both mass failure and undercutting erosion. They found that there was a close relationship between the content of fine-grained particles and soil erosion. The rate of erosion is lowered by an increase in fineness. Majumdar and Mandal (2021) studied on the estimation of river bank erosion risk potential using mechanical and erodibility analysis of soil on the left bank of the Ganga river near Malda district in West Bengal, India. Their results reveal that the erodibility levels increase from high to moderate along an extended left bank line (middle to lower) and fall relatively low along another line on the upper branch of the line according to the 'ROM' scale. They concluded that the nature of soil along left bank is dominantly sandy in which promotes vulnerable condition of river bank sites. They also conclude that basic parameters of soil and its mechanical analysis also reveals that unstable condition exists along river bank line but instability condition is increasing from upper to lower segment of bank line. They recommended that the risk of river bank failure can be measured by determining textural composition of soil. Das and Deka (2020) studied on estimating bank erosion vulnerable areas in Kamrup district, Assam, India using 'ROM' scale. They observed that most of the soil categories in the study area have soils that have a higher proportion of sand and thus have greater risk of bank collapse. They found sandy soil is recognized as minimal cohesive force and thus more susceptible to erosion and transportation by stream. They established by the 'ROM' scale calculation where maximum number of the soil samples was sorted as very high and critical in the degree level of erosion vulnerability and is susceptible to bank failure. They recommended that suitable mitigating methods can be planned for prevention of bank erosion.

The main objective of this study is to categorize the erosion risk potential in terms of particle sizes of bank material.

Methodology

A total of 30 soil samples data has been collected from 6 holes within 6.0 km reach of the left and right bank of Arial Khan River of Bangladesh. The coordinates of the holes of the left bank and right bank have been shown in **Table 2**. The study area and boring layout have been shown in **Fig. 1** and **Fig. 2** respectively. The soil samples have been collected along the holes up to the depth 7.50 m. Here the soil data have been considered up to the depth 7.50 m as the lowest water level near the depth of soil layer. The lowest water level in Arial khan river is (0.13-1.16) mPWD. Particle sizes are analyzed through Laser Diffraction Particle Size Analyzer (Mastersizer 3000). In this study erodibility index has been determined using the $BEI = \frac{\% \text{ Sand} + \% \text{ Silt}}{\% \text{ Clay}}$ and $EI_{ROM} \text{ equation} = \frac{\% \text{ Sand} + \% \text{ Silt}}{2 \times (\% \text{ Clay})}$. Soil erodibility has been categorized using 'ROM' scale.

Table 2. Showing the coordinates of boring points along the left and right bank of Arial Khan River.

Left bank	Coordinate (UTM Zone 45N)		Right bank	Coordinate (UTM Zone 45N)	
	Easting(m)	Northing(m)		Easting(m)	Northing(m)
Khas Char Bachamara	204966.44	2589225.30	Sannyasir Char	204142.34	2590445.94
Char Arial Khan	203086.31	2591543.68	Khankandi	205064.15	2587333.27
Bajehar Char	205717.15	2585987.13	UttarcharTajpur	205419.62	2584340.73

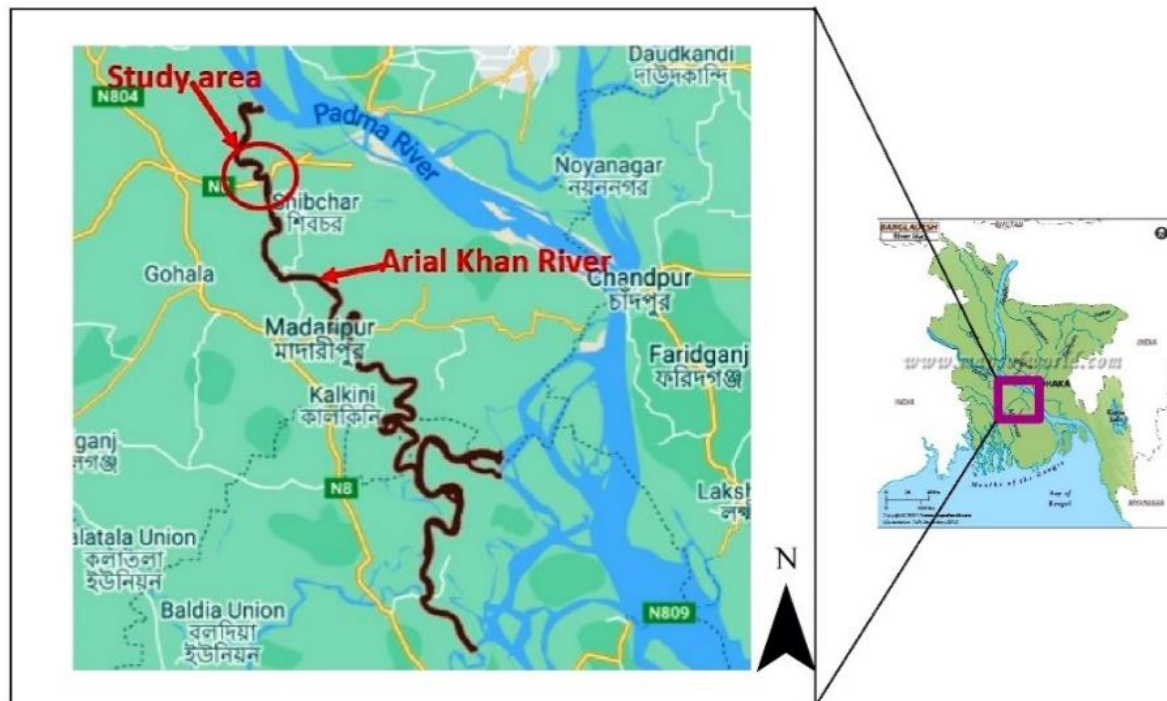


Fig. 1. Study area at and around the Haji Shariotullah bridge.

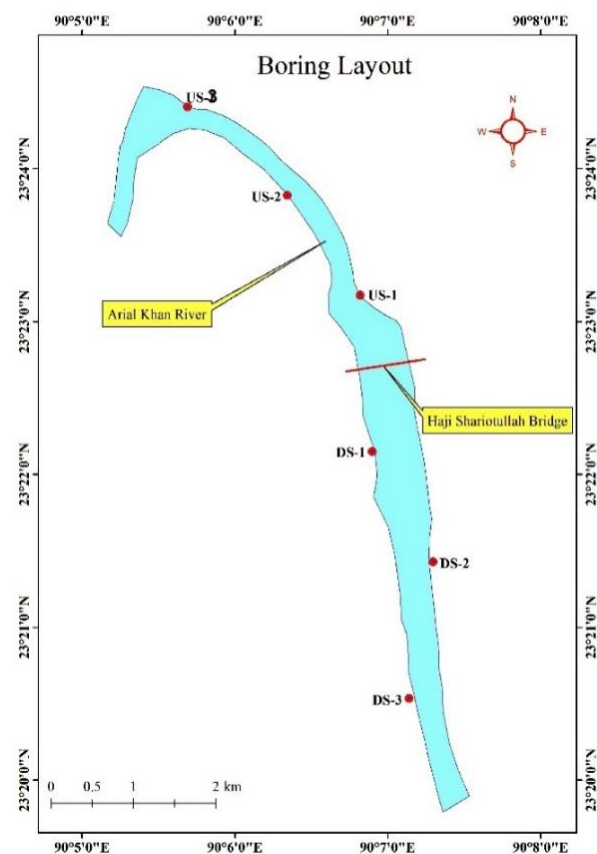


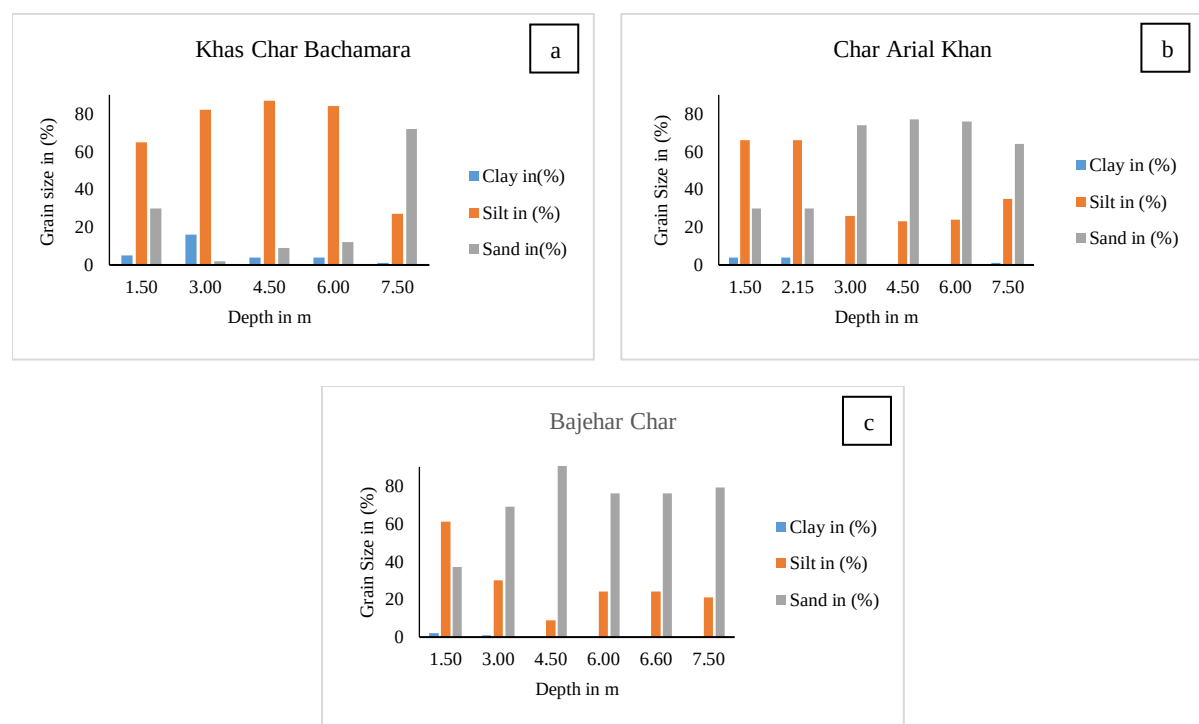
Fig. 2. Boring Layout of the study area.

Results and Discussions

The particle sizes of the soil samples, Bouyoucos erodibility index (BEI) and EI_{ROM} of the left bank of Arial Khan River have been shown in **Table 3**.

Table 3. Showing the percentage of Particle sizes, BEI and EI_{ROM} of the left bank of Arial Khan River.

Left bank of Arial Khan River	Coordinate (UTM Zone 45N)		Depth in m	Particle sizes Analysis			Bouyoucos Erodibility Index (BEI)	EI _{ROM}
	Easting	Northing		Clay in (%)	Silt in (%)	Sand in (%)		
Khas Char Bachamara	204966.44	2589225.30	1.50	5	65	30	19.00	9.50
			3.00	16	82	2	5.25	2.63
			4.50	4	87	9	24.00	12.00
			6.00	4	84	12	24.00	12.00
			7.50	1	27	72	99.00	49.50
Char Arial Khan	203086.31	2591543.68	1.50	4	66	30	24.00	12.00
			3.00	0	26	74	infinity	infinity
			4.50	0	23	77	infinity	infinity
			6.00	0	24	76	infinity	infinity
			7.50	1	35	64	99.00	49.50
Bajehar Char	205717.15	2585987.13	1.50	2	61	37	49.00	24.50
			3.00	1	30	69	9900	49.50
			4.50	0	9	91	infinity	infinity
			6.00	0	24	76	infinity	infinity
			7.50	0	21	79	infinity	infinity

**Fig. 3.** Showing the Particle Size analysis (a) at Khas Char Bachamara (b) at Char Arial Khan (c) at Bajehar Char.

From **Table 3** and **Fig. 3**, it has been observed that Silt particles dominate at Khas Char Bachamara up to the depth of about 6.0 m and then it consists of Sand layer. On the other hand the layer up to the depth 1.5 m of Char Arial Khan

consists of silt layer and then it consists of Sand layer. However, sand particles dominate at Bajehar Char up to the depth of about 7.5 m except upper layer of depth 1.5 m.

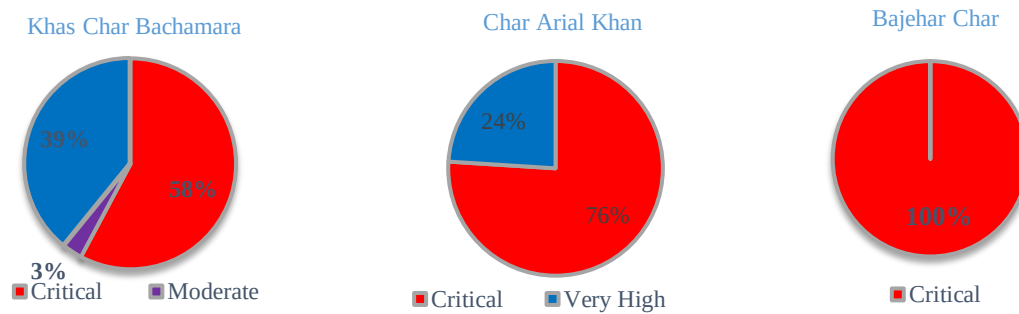


Fig. 4. Illustration of soil erodibility category at Khas Char Bachamara, Char Arial Khan and Bajehar Char.

From **Table 3** and **Fig. 4**, it has been observed that the erodibility index at the bank of Khas Char Bachamara at depth 1.5 m is 9.5 is very high and moderate at depth 3.0 m. However, the erodibility index is moderate to critical at the left bank of Arial Khan River along the bore hole. The erodibility index at the bank of Char Arial Khan is very high to critical at the left bank of Arial Khan River along the bore hole and the erodibility index at the bank of Bajehar Char is

critical along the bore hole. In brief, the left bank of Arial Khan River is very high to critical position in risk of erosion except 3.0 m depth of Khas Char Bachamara which protects the bank from failure for little time. The soil erodibility category in terms of particle sizes has been shown in pie graph in **Fig. 4**. The particle sizes of the soil samples, Bouyoucos erodibility index and E_{IROM} of the right bank of Arial Khan River have been shown in **Table 4**.

Table 4. Showing the percentage of Particle sizes, Bouyoucos erodibility index and E_{IROM} of the right bank of Arial Khan River.

Right bank of Arial Khan River	Coordinate (UTM Zone 45N)		Depth in m	Particle Sizes Analysis			Bouyoucos Erodibility Index (BEI)	E_{IROM}
	Easting	Northing		Clay in (%)	Silt in (%)	Sand in (%)		
Sannyasir Char	204142.34	2590445.94	1.50	4	74	22	24.00	12.00
			3.00	5	65	30	19.00	9.50
			4.50	0	37	63	infinity	infinity
			6.00	0	40	60	infinity	infinity
			7.50	0	34	66	infinity	infinity
Khankandi	205064.15	2587333.27	1.50	0	11	89	infinity	infinity
			3.00	1	70	29	99.00	49.00
			4.50	0	39	61	infinity	infinity
			6.00	0	46	54	infinity	infinity
			7.50	0	30	70	infinity	infinity
Uttarchar Tajpur	205419.62	2584340.74	1.50	1	44	55	99.00	49.50
			3.00	6	86	8	15.67	7.83
			4.50	5	81	14	19.50	9.50
			6.00	3	84	13	32.33	16.17
			7.50	5	81	14	19.00	9.50

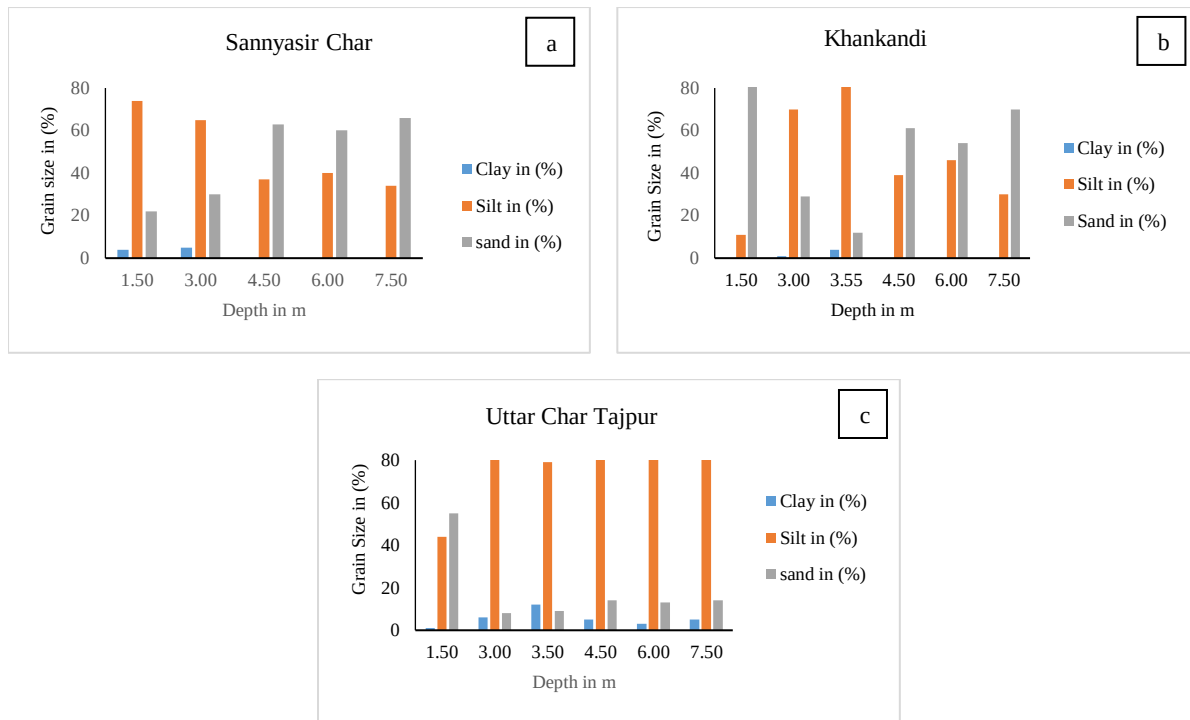


Fig. 5. Showing the Particle Size Analysis (a) at Sannyasir Char (b) at Khankandi (c) at Uttar Char Tajpur.

From **Table 4** and **Fig. 5**, it has been observed that Silt and Sand particles dominate at Sannyasir Char and Khankandi up to the depth of exploration, whereas Silt particle dominates at

Uttar Char Tajpur except the upper layer of about 1.5 m which consists of Silt and Sand layer.

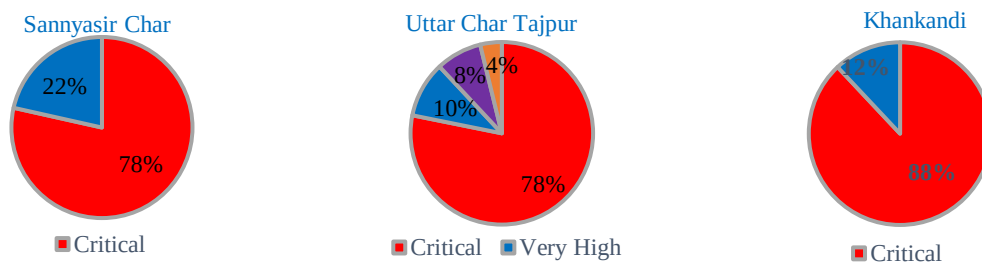


Fig. 6. Illustration of soil erodibility category at Sannyasir Char, Khankandi & Uttar Char Tajpur.

From **Table 4** and **Fig. 6**, it has been observed that the erodibility index at the bank of Sannyasir Char is very high to critical at the right bank of Arial Khan River along the bore hole. The erodibility index at the bank of Khankandi is critical at the right bank of Arial Khan River along the bore hole and the erodibility index at the bank of Uttar Char Tajpur is high to critical along the bore hole. In brief, the right bank of Arial Khan River is high to critical position in risk of erosion.

Conclusions

It has been observed from the left bank of Arial Khan River that Silt particle dominates at Khas Char Bachamara up to the depth of about 6.0 m and then it consists of Sand layer. On the other hand the layer up to the depth 1.5 m of Char Arial Khan consists of Silt layer and then it consists of Sand layer. However, Sand particle dominates at Bajehar Char up to the depth of about 7.5 m except upper layer of depth 1.5 m.

The erodibility index has been observed at the bank of Khas Char Bachamara at depth 7.5 m is critical and moderate at depth 3.0 m. However, the erodibility index is moderate to critical at the left bank of Arial Khan River along the bore hole. The erodibility index at the bank of Char Arial Khan very high to critical at the left bank of Arial Khan River along the bore hole and the erodibility index at the bank of Bajehar Char is critical along the bore hole. In brief, the left bank of Arial Khan River is very high to critical position in risk of erosion except 3.0 m depth of Khas Char Bachamara which protects the bank from failure for little time.

It has been observed from the right bank of Arial Khan River that Silt and Sand particle dominate at Sannyasir Char and Khankandi up to the depth of exploration whereas Silt particle dominates at Uttar Char Tajpur except the upper layer of about 1.5 m which consists of Silt and Sand layer. The erodibility index has been observed at the bank of Sannyasir Char is very high to critical at the right bank of

Arial Khan River along the bore hole. The erodibility index at the bank of Khankandi is critical at the right bank of Arial Khan River along the bore hole and the erodibility index at the bank of Uttar Char Tajpur is high to critical along the bore hole. In brief, the right bank of Arial Khan River is high to critical position in risk of erosion. The study result can help a design Engineer. However, it is recommended here that full geotechnical investigation is needed to prevent the erosion of the river banks.

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